

Project 3: Predicting Cardiac Events based on ECG Data

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Introduction of Dataset

- Contains ECG (electrocardiogram) recordings from 15 different patients, all with congestive heart failure.
 - ECG signals were recorded every .004 seconds over the span of 20 hours
 - Data was subsampled to take the first recording each minute
 - Each signal contains a cardiac event
- Cardiac Events:
 - N – Normal Beat
 - V – Ventricular Ectopic Beat
 - S/R - Premature Beat

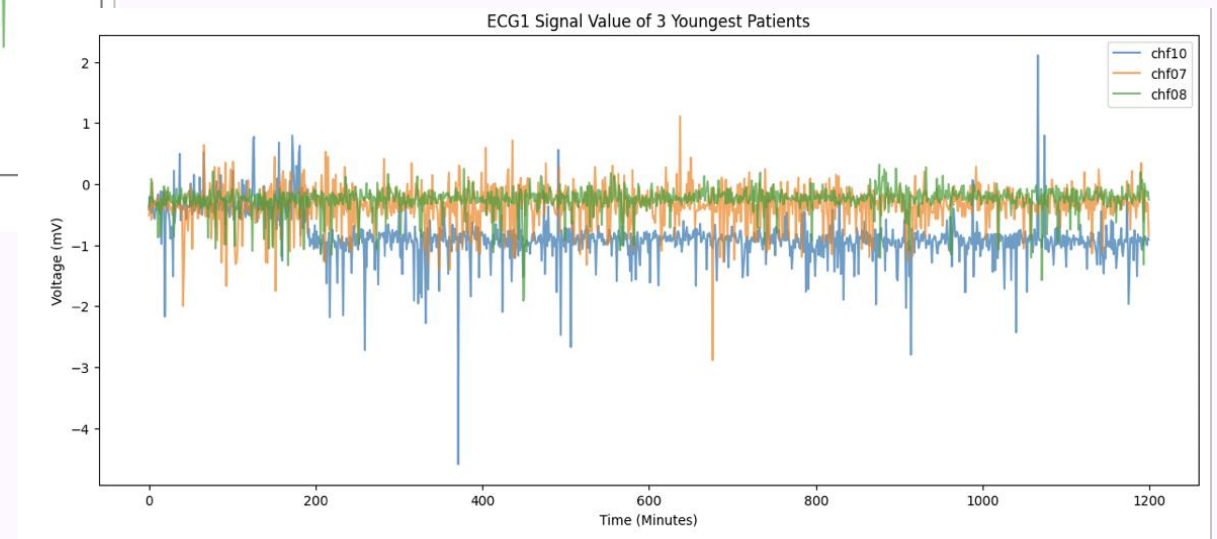
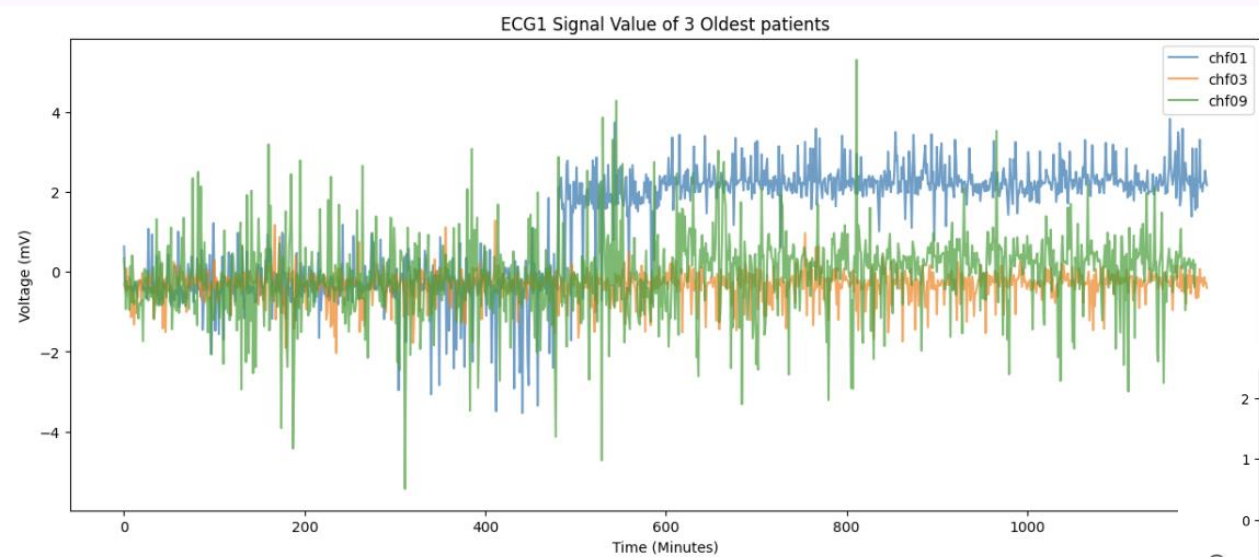
Research Goals

1. Can we train a model to correctly classify the cardiac event based on the patient's ECG data?
2. How can other variables like age affect ECG data?

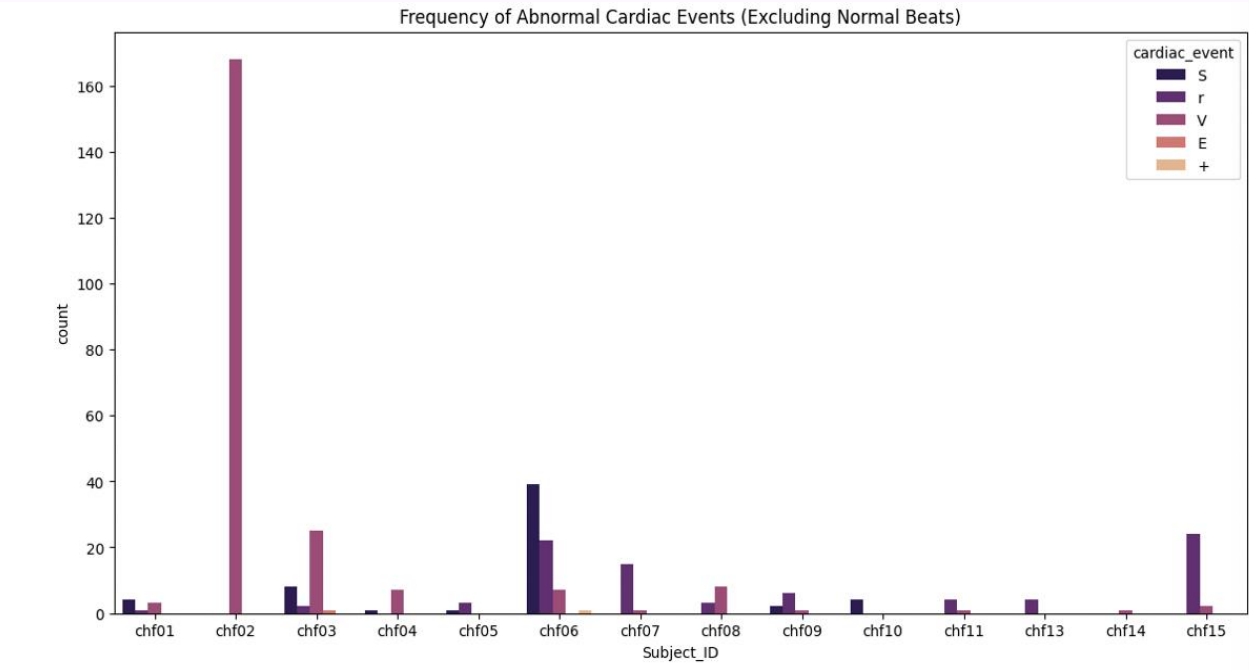
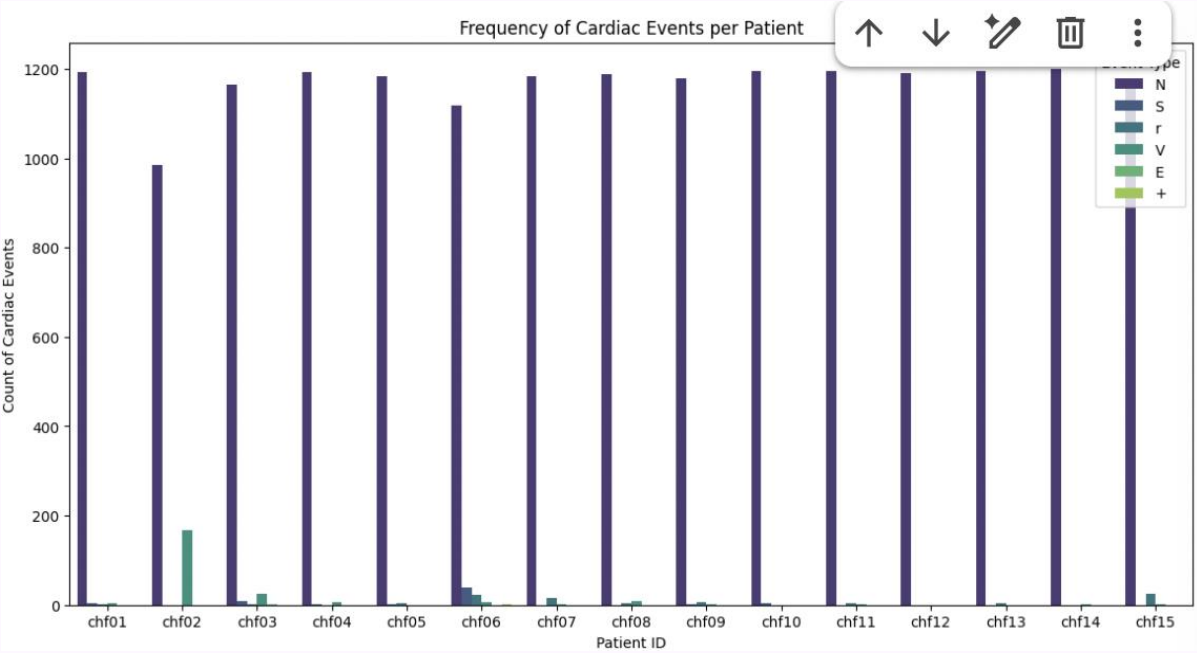
Data Cleaning and Preprocessing

- 20 hour recordings at 250 samples per second required subsampling
- Subsampled by the minute taking one recording from each 60 second interval
- Creating separate Dataframe with annotations to see patient information (age, gender, etc)

Visualization 1: Comparing Age Ranges with ECG data + event



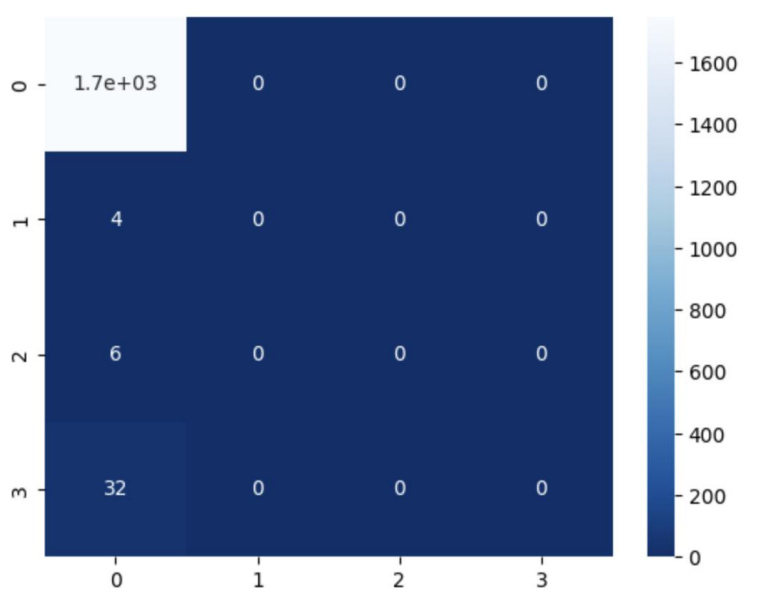
Cardiac Event Counts



ML Model and Evaluation

Target (y) = Cardiac Events

Features (X) = ECG signals 1 and 2



	precision	recall	f1-score	support
0	0.98	1.00	0.99	1749
1	0.00	0.00	0.00	4
2	0.00	0.00	0.00	6
3	0.00	0.00	0.00	32
accuracy			0.98	1791
macro avg	0.24	0.25	0.25	1791
weighted avg	0.95	0.98	0.96	1791

Conclusion/Findings

- Older patients have more volatile ECG readings than younger patients, making them harder to project
- Abnormal cardiac events are very rare, making them difficult to predict
- Predicting normal beats is much easier to predict

Further Steps

- Small number of patients with a low amount of events affects accuracy of model
- Larger dataset with more features
- Comparative data after drug treatment
- Improve model
 - o Use different target

Ethics/Mindfulness

- Real medical patients with heart failure
 - o Private information
 - o Consent needed to use and publicize data
- Possible bias in data
 - o Only 4 women
 - o Only 1 patient below the age of 48
- Transparency that model may not be accurate and should not be completely relied on